

Mama

Contact:

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1. Description of the Overall Project

1.1 Presentation

Mama is building a new production site.

1.2 Background

This greenfield project is currently in the detailed design phase. The AGV system must serve 10 filling lines with packaging material and transport finished goods to a central storage location in the dispatch area. The vehicles will receive packaging material from two warehouses. They will be used indoors in a non-Ex (non-explosive) atmosphere. The current plant design minimizes tracks shared with other traffic like manual forklifts or pedestrians.

1.3 Main tasks of the AGVS

Designation:	A flexible automated guided transport system (AGVS)
Field of application:	Transportation of palletized finished products from the filling area to the dispatch area. Supply packaging material from the warehouse to the filling line.
Basic goal:	Achieve maximum flexibility in work processes and material flows while avoiding line stops due to missing packaging material or backlogs. The system should also avoid the need for manual coordination of multiple forklift trucks for transports.

2. Operating Times

General operating times:	Up to 3 shifts, 3×8 hours per day, up to 7 days per week.
Specifics:	The Hotmelt lines operate 24/7, while other technologies run in 2 shifts from Monday to Friday.

3. Basic Offer Data

3.1 Contents and scope

The offer should be prepared according to a specific structure, with fixed prices and options stated separately. If Mama's resources are used, this must be explicitly mentioned. Consequences for ongoing operations, such as idle times during build-up, commissioning, and acceptance, must be noted.

The offer structure should include:

- A short description of the initial situation and basic requirements.
- A list of laws, standards, and regulations the offer is based on.
- The preparation of a performance specification.
- Details of the AGVS itself, including vehicle layout, dimensions, and power supply.
- A description of how the system will be integrated into the operational environment.
- A plan for commissioning and certification.
- Documentation and a specification of the system's availability and reliability.
- Details on spare parts, maintenance agreements, and other services.
- An estimation of annual energy costs and other costs to be borne by the client.
- An overview of the project team, responsibilities, and schedule.

3.2 Details of the contractor

The contractor must provide a general presentation of their company, including contact persons for the project and a list of reference plants.

4. Assembly, Installation and Commissioning

The contractor must announce assembly, installation, and commissioning activities to Mama at least 4 weeks in advance. This announcement must include a detailed schedule and specify the staff involved. All staff, including subcontractors, must adhere to Mama's site regulations, particularly safety measures. Any use of Mama's resources or alterations to existing facilities must be explicitly authorized.

5. Particularities concerning execution of the project

The overall project is in a detailed engineering phase, which may lead to changes in the layout and tracks. Mama intends to be able to change the vehicle tracks independently, and the AGVS must allow for this. An MES (Manufacturing Execution System) is foreseen for the site, and the interface between the AGVS and MES must be defined during the project.

6. Standards, regulations, laws

The contractor is responsible for observing all binding legal standards and must inform Mama of any deviations. A specification of all valid regulations must be part of the performance specification.

7. Language

Project execution	Standard language for communication will be English.
Software	User interfaces and prints shall be in English or Turkish, and the software source code comments must be in English.
Documentation	Documents must be in English.
Training	Training shall be carried out in English or Turkish.

Possible costs for translations or interpreters must be borne by the contractor.

8. Documentation, Training

The contractor must provide one printed copy of all documents and three copies on data carriers (CD, DVD). Technical drawings must be in DWG format. Operators and maintenance engineers must be properly trained by the contractor.

9. Performance of the System

The target performance is defined by the data and cycle times in Annex 3. The performance P is calculated as the number of transports (n_T) divided by the effective duration of the test (t_{eff}). The AGVS must have a 30% spare capacity to cover future volume increases.

10. Availability, reliability

The overall system or its independent sub-systems must have an availability of at least 98% and a reliability of at least 98%. The condition for each single vehicle is an availability of at least 96% and a reliability of at least 95%. The contractor must present an availability model to determine the plant values.

11. Acceptance of the offered system

The acceptance will be based on the agreed-upon performance specification, Mama's order, and the contractor's binding offer. The acceptance test will be carried out in several steps, including:

- Acceptance of vehicles and single maintenance groups.
- A test of completeness and correctness of the delivery.
- A function test of the plant and its elements.
- Acceptance by an authority.
- A test for coupling and integration with central DP.
- A test operation under real conditions.
- A performance test of the overall system.
- An availability and reliability check, performed at least three months after the first operation.
- A final, formal acceptance.

The final acceptance will transfer risk from the contractor to the client, and the warranty period will begin. If the acceptance test or parts of it fail due to reasons attributable to the contractor, Mama is not obligated to provide resources for a repeated test free of charge.

12. Warranty

The warranty for the contractor's products and services will start upon the transfer of risks.

13. Transfer of risks

The risks for products and services will be transferred upon successful acceptance.

14. List of valid annexes to the tender specification

- Layouts of the hall and equipment.
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Name of handling stations:

ID	Name
P01	Line 01: PSA Line
P02	Line 02: Easymelt Line
P03	Line 04: EVA Line
P04	Line 02-05: Empty pallets
P05	Line 02-05: FG pallets
P06	Line 7: PVC&LSAD
P07	Line 6: WBA Line (Hobbock)
P08	Linie 5: WBA Line 1 (IBC)
P09	Linie 5: WBA Line 2 (IBC)
P10	Line 9: MPT Liq 1 (IBC)
P11	Line 9: MPT Liq 1 (Can)
P12	Line 9: MPT Liq 2 (FG)

ID	Name
P13	Line 9: MPT Liq 2 (IBC)
P14	Warehouse 1
P15	Staging area
P16	Warehouse 2
P17	Charging station
P18	Waiting, Service station
P19	Line 02-04: FG pallets (backup)
P20	Line 05 & 03: Wood Line & Supra Line
P21	Line 7: CS Line (IBC&Can)

Annex 1. Track length and conveyed goods

ID	From	To	Distance (ave) (m)
1	Warehouse 1	Line 01: PSA Line	58
2	Warehouse 1	Line 01: PSA Line	58
3	Line 01: PSA Line	Staging area	82
4	Warehouse 1	Line 02: Easymelt Line	44
5	Warehouse 1	Line 04: EVA Line	23
6	Warehouse 1	Line 05 & 03: Wood Line & Supra Line	60
7	Warehouse 1	Line 02-05: Empty pallets	52
8	Line 02-05: FG pallets	Staging area	17
10	Warehouse 2	Line 7: PVC&LSAD	50

ID	From	To	Distance (ave) (m)
11	Line 7: PVC&LSAD	Staging area	16
12	Warehouse 2	Line 6: WBA Line (Hobbock)	35
13	Warehouse 2	Line 6: WBA Line (Hobbock)	35
14	Line 6: WBA Line (Hobbock)	Staging area	20
15	Warehouse 2	Line 5: WBA Line 1 (IBC)	34
16	Line 5: WBA Line 2 (IBC)	Staging area	21
17	Warehouse 2	Line 9: MPT Liq 1 (IBC)	54
18	Line 9: MPT Liq 1 (IBC)	Staging area	54
19	Warehouse 2	Line 9: MPT Liq 1 (Can)	60
20	Warehouse 2	Line 9: MPT Liq 1 (Can)	60
21	Line 9: MPT Liq 2 (FG)	Staging area	64
22	Warehouse 2	Line 9: MPT Liq 2 (IBC)	66
23	Line 9: MPT Liq 2 (IBC)	Staging area	59
27	Warehouse 2	Line 7: CS Line (IBC&Can)	38
28	Warehouse 2	Line 7: CS Line (IBC&Can)	38
29	Line 7: CS Line (IBC&Can)	Staging area	25

Annex 2. Data of the load handling points

ID	Name	Handling point	Direction of movement	No. of pallets places
P01	Line 01: PSA Line	Floor	delivery & picking	6
P02	Line 02: Easymelt Line	Floor	delivery	6
P03	Line 04: EVA Line	Floor	delivery	6

ID	Name	Handling point	Direction of movement	No. of pallets places
P04	Line 02-05: Empty pallets	Floor	delivery	1
P05	Line 02-05: FG pallets	Conveyor belt	picking	1
P06	Line 7: PVC&LSAD	Floor	delivery & picking	6
P07	Line 6: WBA Line (Hobbock)	Floor	delivery & picking	6
P08	Line 5: WBA Line 1 (IBC)	Floor	delivery & picking	6
P09	Line 5: WBA Line 2 (IBC)	Floor	delivery & picking	6
P10	Line 9: MPT Liq 1 (IBC)	Floor	delivery & picking	6
P11	Line 9: MPT Liq 1 (Can)	Floor	delivery	6
P12	Line 9: MPT Liq 2 (FG)	Floor	picking	6
P13	Line 9: MPT Liq 2 (IBC)	Floor	picking	6
P14	Warehouse 1	Floor	delivery & picking	3
P15	Staging area	Floor	delivery	168
P16	Warehouse 2	Floor	delivery & picking	3
P17	Charging station	Floor		
P18	Waiting, Service station	Floor		
P19	Line 02-04: FG pallets (backup)	Floor	picking	6
P20	Line 05 & 03: Wood Line & Supra Line	Floor	delivery	6
P21	Line 7: CS Line (IBC&Can)	Floor	delivery & picking	6

Annex 3. Cycle time list, transport capacity

ID	Name	Operation time Hour/day	Days/week	Total transports per day
P01	Line 01: PSA Line	16	5	1 (empty pallets), 4 (packing material), 9 (staging area)
P02	Line 02: Easymelt Line	24	7	15 (packing material)
P03	Line 04: EVA Line	24	7	1 (packing material)
P04	Line 02-05: Empty pallets	16	5	13 (empty pallets)
P05	Line 02-05: FG pallets	16	5	128 (staging area)
P06	Line 7: PVC&LSAD	16	5	40 (packing material), 40 (staging area)
P07	Line 6: WBA Line (Hobbock)	16	5	3 (empty pallets), 3 (packing material), 35 (staging area)
P08	Linie 5: WBA Line 1 (IBC)	16	5	9 (packing material), 9 (staging area)
P09	Linie 5: WBA Line 2 (IBC)	16	5	9 (packing material), 9 (staging area)
P10	Line 9: MPT Liq 1 (IBC)	16	5	12 (packing material), 12 (staging area)
P11	Line 9: MPT Liq 1 (Can)	16	5	1 (packing material), 7 (staging area)
P12	Line 9: MPT Liq 2 (FG)	16	5	1 (empty pallets), 8 (staging area)
P13	Line 9: MPT Liq 2 (IBC)	16	5	13 (packing material), 13 (staging area)
P20	Line 05 & 03: Wood Line & Supra Line	16	5	1 (packing material)
P21	Line 7: CS Line (IBC)	16	5	3 (packing material), 3 (staging area)

ID	Name	Operation time Hour/day	Days/week	Total transports per day
P21	Line 7: CS Line (Drum)	16	5	3 (empty pallets), 8 (packing material), 33 (staging area)

On average, 27.5 shipments must be performed per hour.

Annex 4. Communication or data exchange with the AGVS

It is planned to connect the AGV control system with a MES. This interface and data exchange will be specified during the project. The preferred interface phase is OPC UA. Data that could be exchanged includes:

- **Downloading:** BOM, filling order, material.
- **Uploading:** AGV performance, delivery status.

Annex 5. List of important standards and guidelines

The following standards are important for the AGVS and its integration:

- EN 1525 "Safety of industrial trucks, auto guided vehicles and their systems".
- EN ISO 3691-4 "Industrial trucks - Safety requirements and verification".
- 2006/42/EG "Machine Directivities".

Annex 6. List of manufacturers and components preferred by Mama

Component	Supplier 1st choice	Supplier 2nd choice	Supplier 3rd choice
Forklift truck	Jungheinrich	Linde	-
Air dryers	FESTO	ATLAS COPCO	-
AS Interface	IFM	PEPPERL & FUCHS	SIEMENS
Automatic circuit breakers	SIEMENS	ABB	Schneider Electric
Control cabinets	RITTAL	SIEMENS	OMRON
Control panel	SIEMENS	Emerson	-
Cylinders	FESTO	SMC	BOSCH

Component	Supplier 1st choice	Supplier 2nd choice	Supplier 3rd choice
Final switches	SIEMENS	Schneider Electric	SCHMERSAL
Safety barriers	SICK	OMRON	PILZ
Servo drives	SIEMENS	SEW	ELAU (Schneider Electric)

There are additional tables listing preferred manufacturers for a variety of other components.

17. Work loads (cargo) and load handling attachments

The AGV must handle three different pallet sizes with a maximum load of 2000 kg:

- 800×1200
- 1000×1200
- 1200×1200

Conveyed goods	Good 1	Good 2	Good 3	Good 4	Good 5	Good 6	Good 7	Good 8	Good 9	Good 10
Name	Empty Pallet (800×1200)	Empty Pallet (1200×1200)	Cartons empty boxes	Siliconized Boxes	Kraft Bags	PE Bags	PE Hobbock	CS line (Drum-30kg)	CS line (Drum-60kg)	IBC
Length (m)	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Width (m)	0.8	1.2	1.2/0.8	1.2/0.8	1.2/0.8	1.2/0.8	1.2/0.8	1.2/0.8	1.2/0.8	1
Height (m)	1.4	1.5	1	2	1.1	1	2	2	2	1.2
Mass (kg)	170	250	100	100	500	450	225	300	250	60

18. Climate, dusts, fluids, other strains

The AGVS must be able to operate in ambient temperatures ranging from 10-30°C indoors.

19. Safety technology

Each AGV must be equipped with all necessary safety devices, including personal and load protection, emergency stop switches, and visual and acoustic signals. The

contractor is responsible for preparing a risk plan with clearly marked escape routes and danger areas.

20. Operation of the single AGV

Each vehicle must have a lockable main switch. It should also have a diagnosis plug for a diagnosis terminal or remote data transmission. Manual operation is possible via a hand-held control or fixed control elements. A display will show status information and a keyboard will allow for job input.

21. Errors, error messages and troubleshooting

Any event that delays or prevents job execution is considered an error and should be clearly displayed. Errors are classified into categories like "temporary errors or warnings," "remediable errors," and "fatal errors." Error messages on the vehicle terminals must be in plain text and in English or the local language. The control system must document and store at least the last 1,000 error protocols.

22. Control - used hardware

Control components should, as far as possible, be selected from Mama's list of preferred manufacturers (see Annex 6).

23. Maintenance and repair

The contractor must provide specifications for plant repair and maintenance after assembly.

24. Spare parts

A complete list of spare parts must be provided after construction is completed.

25. Documentation

The documentation must include complete and partial layouts, function descriptions, user and maintenance manuals, and all installation and wiring diagrams.

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